STRUCTURAL MODEL REPLICATION · MODEL-ROLE COMPARISON

Qwen everything vs. GPT Sol orchestrator + Qwen workhorse

Same four scenarios · 108 interactions per pipeline · one local-model draw per cell

Pipeline at a glance

	Qwen everything	Sol → Qwen
Planner	Qwen3-8B	GPT Sol
Subjects + measure	Qwen3-8B	Qwen3-8B
Design / study	3 factors × 3 levels 27-cell full factorial	5 factors × 3 levels 27-cell orthogonal array
Valid outcomes	**106 / 108**	**108 / 108** 1 measurement recovered
Runtime	**18.6 min**	**40.2 min**
Prediction result	9 / 12 directions	Broader tests; several mixed patterns
Self-confirmation risk	High same model plans, acts, measures	Lower planner separated from subjects
Main weakness	Simple / ambiguous treatments	Qwen rule compliance

3→5 **Richer designs**
more factors, outcomes, placebos

≠ **Not cleaner execution**
auction exposed workhorse limits

Results by setting

Study	Qwen everything
☕ Mug	Outcome: negotiated pri opening offer ✓ · budget
⚖ Bail	Severity, priors, evidenc 3 / 3 clear
💼 Hiring	Knowledge, clarity, confi 3 / 3 clear · 25 / 27 valid
🖼 Auction	Risk, starting price, strat 2 / 3 clear · risk ✕
Overall	Fast, cheap, self-consist narrower science · same-

Bottom line: GPT Sol improved what was tested; Qwen still determined whether the simulated evidence was behaviorally credible.